

Perceptions about Agility Enablers and their Relative Importance: a Collation

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Abstract

As software houses increasingly align themselves to Agile Software Development paradigm, the issue of what constitutes agility is coming to the forefront. Is it enough to follow the rituals and practices of some agile method to deliver the promises of the paradigm? Or, certain enablers need to be in place to prepare the context for the methods to deliver? What are these attributes of agility and what are their relative contributions? This paper intends to initiate a debate on this issue by collating perceptions of agile researchers and practitioners. For this purpose, we administered Delphi Method separately on two panels to see how the researchers and the practitioners perceive the list of enablers, and their relative importance. It is intriguing to note that while the two panels have similar perceptions about most of the enablers, their views differ substantially regarding others. Further investigation into these seeming differences is called for.

Keywords

Agile Software Development, Agility Enablers, Delphi Method, Agile Methods

Introduction

Software houses are increasingly aligning themselves to agile software paradigm because of its perceived capability to respond faster to changing business requirements of the customers. The expected benefits are better compliance with customer requirements, increased flexibility in development, and faster time to market, among others. As more and more software houses and software teams embrace agility, or decide to do so, the question that naturally comes up is what it takes to be 'agile' so as to be true to the promises? Is it enough to pick up one of the agile methods – Scrum, XP, FDD, DSDM, being a few of them – and expect the benefits to accrue automatically? Or, is there something more to it? Are there certain other facets of agility which, if appropriately factored in, might help the methods to deliver better results? Doubts have already been raised about the consensus among practitioners and academicians regarding the concept of 'agility' or 'agile methods' (Conboy 2009), and how much of it can be delivered just by adopting the methods alone? "Unfortunately, agility seems to provide quite a complicated definitional basis for agile methods" (Iivari and Iivari 2011).

The question then becomes, which attributes, if present, would provide a firm with a better platform to transform its capabilities into enhanced agility? This set of attributes can be called *agility enablers* as the firms can possibly leverage these attributes to become more agile. Agile development paradigm, being more a culture than a method, needs these enablers – with due emphases – to be in place to mitigate the risks associated with different sources of uncertainties. Unless there is this understanding of what influences the agile capabilities of a firm, "[a]ll the agile-flavored tools in the world won't make your teams nimble" (Wittman 2014).

This paper aims at initiating a debate with a goal to identify the set of agile enablers, along with their relative importance. We intend to set the ball rolling with discussions on the outcomes of our study. We used Delphi method to see how the agile practitioners perceive the relative contribution of the potential enablers towards agility. We repeated the same exercise on a panel of agile researchers to see if there is

any difference in perception between these two groups regarding the list of enablers and their relative importance. In other words, we wanted to check if ‘Agility in theory’ is in sync with ‘Agility in practice’. While the two groups concurred on the list of enablers, we were intrigued by certain differences in perceptions. Some of these differences, if true, might need to be deliberated upon if we expect academic research initiatives to enrich agile software development practices.

Next section discusses the motivation for pursuing the study. In subsequent sections, we discuss the methodology adopted for the study, the set of enablers identified, the details of the Delphi method, and the outcome of the analysis. We conclude the paper with a plan on how we intend to take the study forward.

Motivation of the Study

The concept of agility in the software world is presumably borrowed from the world of manufacturing. In the manufacturing world, agility was introduced to mitigate the risks of (a) supplier uncertainty, (b) demand uncertainty, and (c) operational uncertainty. The traditional software development methods have also been beleaguered with similar risk factors: (a) *People-related*, in terms of timely availability of human resources with requisite skills and experience, (b) *Customer-related*, in terms of changing needs of the customer, and (c) *Process-related*, in terms of the effectiveness of the processes followed during software development (Poonacha and Bhattacharya 2012).

In an effort to bring out software development from its entrenched practices and, in the process, mitigate the above risks, ‘Manifesto for Agile Software Development’ was brought out in 2001, along with its associated twelve principles (Agile Alliance 2001). These twelve principles can be interpreted as *foundations* of agility rather than the *means* to achieve it. Practitioners, based on their own understanding of these principles and their own work environment, came up with different stencils to become agile. Increasing adoption by different firms and subsequent evolution of agile practices led to a plethora of agile software development methods – XP, FDD, DSDM, Scrum being a few of them. Existence of this many methods and their variants have really complicated the scenario as “[a]gility as a concept... has been used by many different people to refer to very different phenomena” (Conboy 2009). It may be noted that the development of different agile methods has mostly been driven by the practitioners. Until recently, limited scholarly investigation has been undertaken by the academic research community in this area. But with growing adoption of agile practices, the area, in recent times, has been attracting increasing number of academic research initiatives.

Do these two communities – researchers and practitioners – have the same perception about the attributes that facilitate agility? Existence of a multitude of methods, coupled with the lack of cumulative research tradition in the area, may have led the researchers to develop different understanding and appreciation of the principles guiding the agile practices than the practitioners. But to be able to make valuable contributions to the practice of agile software development, it is crucial that both the communities are on the same platform regarding their views on the importance of the agility enablers and their conceptual underpinnings.

A debate in this matter would not only point out conceptual differences, if any, but would also lead to an improved understanding about the interactional dynamics of different enablers. On one hand, any such improved understanding would help the practitioners to make a better decision in choosing a method in a certain context; on the other hand, it would allow the practitioners and researchers to fine-tune the methods to become more agile. This research is not the debate itself but an initiation of it.

Methodology Used

To bootstrap the process, we first surveyed literature on agile software development to prepare the initial list of possible agility enablers. Once the list was made, the list, along with succinct explanations of each, was floated among a group of agile researchers and, separately, among a group of agile practitioners as the initial suggestion for agility enablers.

We used Delphi method to collate perceptions regarding the enablers, and their relative importance. It may be recalled that, since 1950, Delphi method has been employed by number of researchers for obtaining “the most reliable consensus of opinion of a group of experts” (Dalkey and Helmer 1963).

Delphi approach provides the opportunity to the participants to revise their own views on the basis of comments provided by others. Anonymity was maintained in both the panels.

We invited agile researchers and practitioners, satisfying pre-set criteria, to be members of the respective panels. Delphi method was administered separately for the two panels. Inputs were collected through structured spreadsheets, exchanged through emails. We stopped after third iteration for reasons to be explained later. The data collected till that point was analyzed to create the prioritized list of the enablers.

We then used Wilcoxon Rank Sum Test (Bradley and South 1981) to see if the importance assigned to each enabler by the two panels were coming from the same distribution.

Set of Enablers

As mentioned before, agility enablers can be thought of as the set of “enabling factors and conditions, some external to the actual software development function, [which] must be fulfilled” (Kettunen and Laanti 2008) to achieve enhanced agility.

From extant literature about agile software development, we identified a list of papers dealing with attributes/factors/conditions that influence the outcomes of agile practices. From this set of papers, we then tried to identify the set of attributes/factors which have been found by the researchers to have significant contribution in preparing the context that would facilitate agility. These attributes/factors help in neutralizing different sources of uncertainties through effective handling of the issues that the practitioners are seized with for years. These attributes were pulled together to form the initial list of sub-constructs for agility enablers. The attributes were then mapped to the set of Agile Principles (Agile Alliance 2001). The rationale used for this mapping is being omitted here because of paucity of space. Based on this mapping, redundancies in the list of attributes were identified and the list was accordingly trimmed down. From this mapping, attributes that help in managing similar issues and accrue similar benefits to the firm were identified. On the basis of these similarities, attributes were clustered, and accordingly named. Taking cues from the literature, we also came up with short explanatory comment on each of the clusters. The outcome was a list of 11 proposed enablers.

Table 1 lists the enablers, as was finalized after the Delphi method with some renaming of the enablers, and minor tweaking of the interpretations of what was suggested initially. To keep the number of references under control, we have cited here the smallest set of references which, together, cover discussions on all the enablers and their explanations.

1	Enabler: Team Capability	Associated Agile Principles: 5, 7, 8	Literature Reference(s): (Misra et al. 2009; Poonacha and Bhattacharya 2012)
...talks about the competencies of the team members associated in a project. It can be enhanced by ensuring functional flexibility in a team. Functional flexibility can be achieved through deploying multi-skilled workforce that would facilitate cross-functional deployment.			
2	Enabler: Self-Organizing Team	Associated Agile Principles: 5, 11	Literature Reference(s): (Alaa and Fitzgerald 2013; Misra et al. 2009)
...is made of motivated individuals who, not only execute self-assigned tasks, but are empowered to exercise their professional judgment to decide what would be best for the success of the project. Internalized plans and qualitative controls are prepared, deployed and monitored by the team members rather than external managers. Self-Organizing Teams gives team members more space to take decisions without consulting upper management.			
3	Enabler: Knowledge Sharing and Management	Associated Agile Principles: 6, 12	Literature Reference(s): (Misra et al. 2009; Misra et al. 2010; Williams et al. 2010)
Knowledge management strategies in the agile domain relates to management of tacit knowledge that mostly resides within the agile developers. It is perceived that agile practices require less formal training; mentoring, and professionally guided discussions are more useful in transfer of tacit knowledge between individuals. Knowledge Sharing can improve team competencies and functional			

flexibility of the team members.			
4	Enabler: Organizational/ Team Culture	Associated Agile Principles: 5, 8	Literature Reference(s): (Misra et al. 2009; Misra et al. 2010; Williams et al. 2010)
...assesses the extent to which prevalent culture at a specific organization/team is suited and adaptive to agile principles. Decentralized decision-making, collaborative attitude, sense of responsibility, readiness to learn, customer-centric processes along with adequate management support are considered to be important aspects of the culture of an organization/team that promotes agility.			
5	Enabler: Iterative Planning and Development	Associated Agile Principles: 1, 2, 3, 7	Literature Reference(s): (Alaa and Fitzgerald 2013; Cao and Ramesh 2008; Williams et al. 2010)
Teams following agile methodologies work in short iterations. All the requirements are not often pre-defined; rather they emerge during these iterations. Iterative development ensures continuous delivery of working software at regular intervals. Continuous integration and automated builds are other two important aspects of it. Iterative development helps in easier accommodation of suggested changes.			
6	Enabler: Simplicity in Design and Architecture	Associated Agile Principles: 2, 9, 10	Literature Reference(s): (Alaa and Fitzgerald 2013; Poonacha and Bhattacharya 2012)
...entails using abstractions/metaphors, reusable modules etc. Simple, feature-based design is practiced to enhance the degree of modularity of the code, which, in turn, enhances reusability of the same. Modular practices enhance flexibility since a change in requirement would affect a few modules and would not require a complete system redesign.			
7	Enabler: Continuous Improvement	Associated Agile Principles: 12	Literature Reference(s): (Alaa and Fitzgerald 2013; Poonacha and Bhattacharya 2012; Williams et al. 2010)
...refers to the on-going efforts taken up by an organization to tune its development processes towards better results. It refers to the extent to which the team conducts retrospectives, daily meetings etc. to reflect upon the lessons learned and how to incorporate those going forward.			
8	Enabler: Quality Management	Associated Agile Principles: 9	Literature Reference(s): (Alaa and Fitzgerald 2013; Williams et al. 2010; Williams 2012)
...refers to the efforts the development team explicitly puts in to improve the quality of the software being developed. To ensure higher quality, agile teams adopt different practices like Refactoring, Automated Unit Testing, Coding Standard, Code and Design Inspections, Parallel Testing etc.			
9	Enabler: Face-to-face/Effective Communication	Associated Agile Principles: 4, 6	Literature Reference(s): (Cao and Ramesh 2008; Cao et al 2010 ; Misra et al. 2009)
...aims at faster and effective exchange of information between the development team and the customer. Both frequency and mode of communication are critical determinants of communication effectiveness of an organization/team. On-site Customers, Co-located teams etc. are often recognized as vehicles of Effective Communication.			
10	Enabler: Requirement Capturing and Prioritization	Associated Agile Principles: 2, 3	Literature Reference(s): (Cao and Ramesh 2008; Williams et al. 2010; Williams 2012)
A mature requirement capture process would involve relevant stakeholders and ensure requirements to be verifiable, traceable, and prioritized based on their relative importance. Requirement			

prioritization starts in early stages to ensure important features are delivered as early as possible. Ability to accommodate changing requirements even at the later stages of development is critical.			
11	Enabler: Customer Involvement and Commitment	Associated Agile Principles: 1, 3, 4, 8, 10	Literature Reference(s): (Alaa and Fitzgerald 2013; Cao et al 2010)
Customer involvement in an agile project goes far beyond requirement capturing and prioritization. Collaborative, knowledgeable and committed customers play critical roles in project estimation, quick decision-making, developing acceptance tests and validating results, iterations and release planning, evaluating product prototypes etc.			

Table 1. List of Enablers

Depending on the type of uncertainties expected to be mitigated by each of the enablers, the identified enablers can be categorized in three groups – People-related, Process-related and Customer-related – as given in Table 2. This categorization was not shared with the panelists. This categorization would help us later in comparing the perceptions of the two groups of respondents at the category-level.

Administering the Delphi Method

As mentioned earlier, we decided to collect perceptions regarding agile enablers from two groups of experts – agile researchers and agile practitioners. While approaching the researchers the criterion was that (s)he must have at least 2 publications in peer-reviewed journals in last 5 years on topics related to agile software development. The criterion was chosen to ensure that the researcher is familiar with issues that are currently being discussed in agile development. As regards practitioners, the criteria was (s)he must have at least 4 years of agile software development experience, and must be in a lead position in his/her organization(s). We also had to ensure that the researchers and the practitioners would be available for the iterations of the Delphi method estimated to take about a month's time.

People-related Enablers	Process-related Enablers	Customer-related Enablers
1. Team Capability	1. Iterative Planning and Development	1. Face-to-face/Effective Communication
2. Self-Organizing Team	2. Simplicity in Design and Architecture	2. Requirement Capturing and Prioritization
3. Knowledge Sharing and Management	3. Continuous Improvement	3. Customer Involvement and Commitment
4. Organizational/Team Culture	4. Quality Management	

Table 2. Categorization of the Enablers

Based on the interest shown and availability committed, we started the process with 11 experts in each panel. The panels were deliberately kept small to facilitate convergence during Delphi method. In the researchers' panel we had seven experts from US, two from Australia and one each from Europe and Asia. In addition to meeting the publication-related criterion, most of them also had experience in teaching courses related to agile software development. In the practitioners' panel, six were from IT services firms, three from IT product firms and the other two had experience in both types of firms. Most of them were in multi-national companies. Majority of them had substantial experience in traditional software development methodology also. The details of the process given below apply to both the panels.

Before initiating the first round, a warm up mail was sent to the panelists to familiarize them with the rules of the game, and the proposed enablers and the interpretations thereof. Each of them was requested to study the enablers, and be prepared to add to, delete from, or alter the list. In the first round, the panelists were asked whether any modification was needed either in the list or in the interpretations. To capture the perceptions about the relative importance of the enablers, the panelists were asked to distribute 100 points among the enablers based on their perceived criticality. Each panelist was supposed to provide a crisp rationale for each of the allocations made, or changes suggested. In subsequent rounds, a summary of the allocations made and comments given by the peer panelists in the previous round was

circulated, maintaining anonymity of the panelists. The panelists were then asked whether, based on the observations of others, they would like to revise their earlier allocations. During the iterations, one expert in each panel dropped off as they found it difficult to respond within the specified time window.

We stopped at the end of the third round as it became clear, from the changes in allocations, and fresh comments, or the lack of it, that not much of a revision of the allocations could be expected in the subsequent rounds. Neither did we expect a complete agreement in allocations in either panel. In the researchers' panel, individuals had leanings on their area(s) of interest as was evident from the match between their emphases and their publications. In the practitioners' panel, their allocations possibly reflected the scope of their experience – small versus large project, collocated versus distributed team, island of agility versus organization-wide agility etc.

Analysis and Observations

Fortunately, no major changes were suggested in the initial list of enablers that was circulated among the panelists. However, there were certain suggestions regarding the names of enablers and wordings of the interpretations. Based on the comments received from the panelists, 'Organization Culture', which was one of the enablers in the original list, was subsequently renamed as 'Organizational/Team Culture' as in various contexts 'Team Culture' can assume more importance than 'Organizational Culture'. Similarly, 'Training and Knowledge Management' was renamed as 'Knowledge Sharing and Management' as it was pointed out that an agile project would not generally involve conventional 'training sessions'; rather it would foster a climate for exchange of knowledge and learning. A few adjustments were also made to the enabler interpretations based on the comments received from the panelists.

A few other changes were also suggested regarding the scope of the enablers but we did not incorporate those as the respondents, as a group, could not agree on the suggestions made. But these suggestions – for example, 'Iterative Planning and Development' subsumes both 'Continuous Improvement' and 'Quality Management', or 'Customer involvement and Commitment' incorporates 'Requirements Capturing and Prioritization' etc. - vindicated the categorization we made on the basis of the sources of uncertainties. Most of the suggestions involved enablers belonging to same categories. It may be recalled that panelists were not informed about the categorization of the enablers during the course of the study.

Enabler-wise Summary of Allocations

Enabler-wise summary of the allocations made by the two panels is given in Table 3. The table also provides priorities of the enablers based on their relative importance as perceived by each panel. Mean and standard deviation have been used as the measures of priority. In the next section we will do Wilcoxon Rank Sum Test to check if the perceptions of the two panels are coming from the same distribution. In each panel, inputs from one panelist were left out from consideration since, from the beginning, these were outliers by far, and did not change much with iterations.

Priorities as perceived Agile Researchers				Priorities as perceived by Agile Practitioners			
Rank	Enabler	Mean	Std Dev	Rank	Enabler	Mean	Std Dev
1	Customer Involvement and Commitment	12.22	2.48	1	Requirement Capturing and Prioritization	12.67	3.23
2	Organizational/Team Culture	11.22	5.05	2	Iterative Planning and Development	11.33	2.31
3	Self-Organizing Team	10.89	1.59	3	Customer Involvement and Commitment	11.22	1.69
4	Face-to-Face/Effective Communication	10.89	2.33	4	Face-to-Face/Effective Communication	10.78	3.68

5	Iterative Planning and Development	10.33	1.89	5	Self-Organizing Team	10.67	1.76
6	Team Capability	8.44	2.41	6	Continuous Improvement	9.56	3.13
7	Continuous Improvement	8.11	1.91	7	Team Capability	8.56	2.59
8	Simplicity in Design and Architecture	7.56	1.71	8	Simplicity in Design and Architecture	7.78	3.33
9	Requirement Capturing and Prioritization	7.22	2.62	9	Organizational/ Team Culture	6.22	2.04
10	Quality Management	6.67	3.09	10	Quality Management	6.22	2.39
11	Knowledge Sharing and Management	6.44	2.67	11	Knowledge Sharing and Management	5.00	2.36

Table 3: Enabler-wise Summary of Allocations

It is interesting to note that while the two panels have similar views regarding criticality of some of the enablers, they have different perceptions regarding importance of others. Based on the data in Table 3, and comments received from the panelists, certain observations are in order:

- (a) The researchers and the practitioners were in sync regarding their perceptions about ‘Face-to-Face/Effective Communication’ and ‘Simplicity in Design and Architecture’ – both in relative importance and in contributed value.
- (b) However, perception regarding ‘Organizational/Team Culture’ varied widely between the two panels. Researchers considered it to be one of the most critical enabler of agility. In contrast, the practitioners considered the relative importance of this enabler to be pretty low – ninth in their list – with lesser variations in allocations among them. One representative comment from researchers was: *“Organizational culture can facilitate or hinder adoption of agile practices so this is the most significant enabler as well as the most significant inhibitor for a successful agile project”*. Despite significant variation in the allocations made to it by the researchers, overall, they considered the relative importance of it to be second only to ‘Customer Involvement and Commitment’. On the other hand, practitioners in general were of the opinion that ‘Organization/ Team Culture’ is important but is not something critical in defining agility – *“It may influence the agile practice but it’s not a core enabler for the practice”*.
- (c) Similar difference in perception can be noticed for ‘Requirement Capturing and Prioritization’ as well. Practitioners’ panel considered it to be the most critical enabler of agility. The following comment by one practitioner sums up the majority view expressed by their panel - *“More often than not, software projects fail because of lack of clarity on the requirements and product management’s inability to prioritize the features...Agile methodologies allow teams to be flexible to changing requirements, but [if] the requirements are not correctly capture[d] and prioritised, agility in the team makes little difference”*.

On the other hand, researchers in general were not ready to give much importance to ‘Requirement Capturing and Prioritization’ as a critical enabler. *“[While] I recognize the importance of requirements capturing and prioritization, I don’t think it is a determinant of agility. Further, this is a critical step for any project, agile or otherwise”*. They, as a group, assigned it to the ninth position in their priority list. Some of them even mentioned that the development team does not need to think about ‘Requirements capturing and prioritization’ as the basic philosophies behind it are already taken care of by other enablers such as ‘Iterative Planning and Development’, ‘Customer Involvement and Commitment’ etc.

- (d) Both panels perceived ‘Quality Management’ and ‘Knowledge Sharing and Management’ to be of least importance in facilitating agility. According to one researcher, “[*Quality Management is*] supposed to be built into the agile process” and hence, need not be considered as an independent enabler. The same perception was echoed by the practitioners as well: “[it to be] *less important than the [other enablers], considering that agile processes by default target to increase the quality*”. Comments like “[Knowledge Sharing and Management] *are not peculiar to agile; neither are they by themselves enablers of agility*” and “[t]hings change very quickly, and having an effective knowledge management is neither possible nor necessary” summarize the perceptions of both the panels regarding ‘Knowledge Sharing and Management’.

Some of these observations – if supported by subsequent studies – are quite intriguing. We are particularly worried about the following:

- (i) Why are the researchers and the practitioners so divergent in their perceptions regarding importance of ‘Organizational/Team Culture’ and ‘Requirement Capturing and Prioritization’ as agility enablers? A quick look at the explanations of each of these would reveal that each had, arguably, little room to be misinterpreted. Even if they were interpreted differently, why should that be consistent within a panel and different across? If true, differences in perceptions of these two enablers need to be resolved if both the groups intend to have same understanding about agile software development.
- (ii) Is quality management so deeply embedded in the practices and rituals of agile methods that it need not be considered separately? Can the same be said about knowledge management? Is it possible that it is not getting its due importance since it does not have any immediate impact on a project? In that case why researchers also are not flagging the importance of it? Or, has ‘Knowledge sharing and management’ become irrelevant in agile context?

Enabler-wise Wilcoxon Rank Sum Test

How confident can we be about these observations? The small sample size and the high standard deviations create some room for doubt. Can we conclude with enough certainty that the panels differ significantly in their perceptions regarding certain enablers? We applied two-tailed Wilcoxon rank sum test (for small independent samples) on the allocation data to check if the perceptions of the two panels were coming from the same distribution. We used the ranksum() function of MATLAB for the purpose. For each enabler, the null hypothesis was that both the panels perceived the enabler to be of similar importance. As shown in Table 4, we could reject the null hypothesis at 10% significance level, for each of ‘Organization/Team Culture’ and ‘Requirement Capturing and Prioritization’, the observed p-values being 0.0030 and 0.0014 respectively. Hence, our observations regarding these two enablers in the previous section can be considered as valid. The null hypotheses could not be rejected for any of the other nine enablers even at 5% level.

No.	Enabler	Observed P Value	Null Hypothesis rejected? Level of significance: 0.1
1	Face-to-face/Effective Communication	0.6652	No
2	Team Capability	0.9888	No
3	Self-Organizing Team	0.5082	No
4	Customer Involvement and Commitment	0.5148	No
5	Requirement Capturing and Prioritization	0.0014	Yes
6	Iterative Planning and Development	0.5019	No
7	Simplicity in Design and Architecture	0.6422	No
8	Continuous Improvement	0.2662	No

9	Training and Knowledge Management Initiative	0.3758	No
10	Quality Management Initiatives	0.6498	No
11	Organizational Culture	0.0030	Yes

Table 4: Wilcoxon Rank Sum Test Results on Enabler-wise Allocations**Category-wise Allocation Summary**

Table 5 provides summary allocations made for the three categories we had defined earlier, i.e., People-related, Process-related and Customer-related. The allocations made to all the enablers belonging to a certain category were summed up to get an idea about the perceptions regarding category-wise relative importance. It may be noted that Customer-related category has 3 enablers in it, while both Process-related and People-related categories have 4 enablers each.

Interestingly, both the panels had assigned almost similar importance to all the three categories, even without knowing about this categorization. From the difference in relative positioning of the three categories in the table, it can be observed that academicians and practitioners perceive the criticality of different sources of uncertainties associated with software development differently. Wilcoxon Rank Sum Test results (not reported here) on category-wise data of individual panelists confirmed that the two panels have different perceptions about the importance of ‘People-related Enablers’ and ‘Customer-related Enablers’.

Ranking as per Agile Researchers				Ranking as per Agile Practitioners			
Rank	Enabler	Mean	Std Dev	Rank	Enabler	Mean	Std Dev
1	People-related Enablers	37.00	5.27	1	Customer-related Enablers	35.00	5.48
2	Process-related Enablers	32.67	3.92	2	Process-related Enablers	34.75	6.70
3	Customer-related Enablers	30.33	3.33	3	People-related Enablers	30.25	4.52

Table 5: Category-wise Summary of Allocations**Moving Forward**

Comparison of perceptions reported here is outcome of a limited study, involving only a few people to facilitate smooth administration of Delphi method. However, the results indicate that there might be a need to initiate a larger debate on agility enablers and their relative importance. The starting point could be the list of identified enablers and their relative contributions to agility as suggested by the panels in this study. The agile researchers and practitioners must be in sync to get the desired outcome of the agile software development paradigm.

We are currently in the process of examining the same issue of relative importance by collecting data on pair-wise comparison of the enablers by other sets of researchers and practitioners. We intend to analyze the data, thus collected, using Analytic Hierarchy Process (AHP) to see if it leads to the same result as obtained by the Delphi method.

As indicated earlier, some of the comments provided by the participants suggest that individuals might be having different understandings about interdependencies among the enablers. A careful scrutiny of the enablers and their interpretations would reveal that there might be sub-constructs for each enabler, and these sub-constructs might have relationships across enablers. These implied partial relationships might be responsible for the differences in opinions, if any, of the macro enablers. In a subsequent phase of the project, we intend to study the causal relationship among the enablers and their sub-constructs. This

analysis, coupled with an understanding of the relative importance of the enablers, would have the potential to lead us to an agility maturity model. This would also possibly help in objectively assessing the agility of a software development organization.

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References

- Agile Alliance. 2001. "Manifesto for Agile Software Development", <http://agilemanifesto.org/>
- Alaa, G., and Fitzgerald, G. 2013. "Re-conceptualizing agile information systems development using complex adaptive systems theory," *Emergence: Complexity & Organization* (15:3), pp. 1-23.
- Bradley, I. E., and South, J. B. 1981. *Introductory Statistics for Business and Economics*, Dryden Press, pp. 452.
- Cao, L., and Ramesh, B. 2008. "Agile requirements engineering practices: An empirical study," *Software, IEEE* (25:1), pp. 60-67.
- Cao, L., Ramesh, B., and Abdel-Hamid, T. 2010. "Modeling dynamics in agile software development," *ACM Transactions on Management Information Systems (TMIS)* (1:1), pp. 5:1-5:26.
- Conboy, K. 2009. "Agility from first principles: reconstructing the concept of agility in information systems development," *Information Systems Research* (20:3), pp. 329-354.
- Dalkey, N., and Helmer, O. 1963. "An experimental application of the Delphi method to the use of experts," *Management science* (9:3), pp. 458-467.
- Iivari, J., and Iivari, N. 2011. "The relationship between organizational culture and the deployment of agile methods," *Information and Software Technology* (53:5), pp. 509-520.
- Kettunen, P., and Laanti, M. 2008. "Combining agile software projects and large-scale organizational agility," *Software Process: Improvement and Practice* (13:2), pp. 183-193.
- Misra, S. C., Kumar, V., and Kumar, U. 2009. "Identifying some important success factors in adopting agile software development practices," *Journal of Systems and Software* (82:11), pp. 1869-1890.
- Misra, S. C., Kumar, V., and Kumar, U. 2010. "Identifying some critical changes required in adopting agile practices in traditional software development projects," *International Journal of Quality & Reliability Management* (27:4), pp. 451-474.
- Poonacha, K. M., and Bhattacharya, S. 2012. "Towards a Framework for Assessing Agility," in *Proceedings of the 45th Annual Hawaii International Conference on System Sciences (HICSS)*, Hawaii, pp. 5329-5338.
- Williams, L., Rubin, K., and Cohn, M. 2010. "Driving process improvement via comparative agility assessment," In *Agile Conference (AGILE), 2010*, IEEE, pp. 3-10.
- Williams, L. 2012. "What agile teams think of agile principles," *Communications of the ACM* (55:4), pp. 71-76.
- Wittman, E. 2014. "What every company should know about agile software development," *MIT Technology Review*, accessed on August 6, 2014, <http://m.technologyreview.com/view/528651/>.